Assignment No 3

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Title – Implement A* Search for 8 puzzle problem.

Program:-

```
import heapq


# Goal state
goal = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)


# Heuristic function: Manhattan Distance
def h(s):
    d = 0
    for i in range(9):
        if s[i] != 0:
            j = goal.index(s[i])


            d += abs(i // 3 - j // 3) + abs(i % 3 - j % 3)


    return d
```

```
# A* Search Algorithm
def astar(start):

    # Priority queue: (f, g, state, path)
    q = [(h(start), 0, start, [])]

    # Set of visited states
    seen = set()

    while q:

        # Get state with minimum f value
        f, g, s, path = heapq.heappop(q)

        # Skip if already visited
        if s in seen:
            continue

        # Mark state as visited
        seen.add(s)

        # Check whether goal is reached
        if s == goal:
```

```
    return path + [s]

# Find position of blank (0)
z = s.index(0)

# Convert position into row and column
r, c = divmod(z, 3)

# Possible moves: Down, Up, Right, Left
for dr, dc in [(1, 0), (-1, 0), (0, 1), (0, -1)]:

    nr, nc = r + dr, c + dc

    # Check whether the move is valid
    if 0 <= nr < 3 and 0 <= nc < 3:

        # Find new position
        x = nr * 3 + nc

        # Create new state
        t = list(s)

        # Swap blank with adjacent tile
```

```
            t[z], t[x] = t[x], t[z]

            # Convert list back to tuple
            t = tuple(t)

            # Calculate f = g + h and add to queue
            heapq.heappush(
                q,
                (g + 1 + h(t), g + 1, t, path + [s])
            )

# Initial state
start = (1, 2, 3,
         4, 0, 6,
         7, 5, 8)

# Find solution
solution = astar(start)

# Display solution
```

```
print("Solution:")
print()

for s in solution:
    print(s[0:3])
    print(s[3:6])
    print(s[6:9])
    print()
```

Output:-

```
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)


Process finished with exit code 0
```